



Geospatial Analysis of a Rip Current Dye Release at Long Beach, British Columbia

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Abstract

Rip currents represent one of the most dangerous and visually inconspicuous hazards at surf beaches. Drone technology is transforming methods for tracking and communicating their flow structure. To assess rip current dynamics fluorescent dye was released into a known headland rip current system and tracked using oblique aerial drone video. Still images were georeferenced to create digitized dye polygons for rip current analysis, including dye plume size and rate of change. An interactive web map was developed to display dye plume polygons through time, enabling visualization of the flow sequence. Results indicated multidirectional initial dispersion within the surf zone, followed by progressive channelization and accelerated transport toward the headland, consistent with headland rip current behaviour. Methods demonstrate the utility of using drone imagery to trace dye as a geospatial tool for rip currents, while exposing limitations inherent to oblique imagery georeferencing, manual digitization, and limited ground control points.

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1. Introduction

Rip currents are narrow, offshore directed flows forming within the surf zone capable of exceeding velocities of 1 m/s and represent one of the leading causes of global coastal drowning deaths (Brander et al., 2014; Dalrymple et al., 2011). In the United States alone, approximately 100 rip current fatalities are recorded annually, a toll exceeding that of floods, hurricanes, and tornadoes combined (Brander et al., 2014). Despite their severity, rip currents receive relatively little attention and funding compared to other natural hazards due to their limited surface visibility and scattered fatalities.

Surveys of beachgoers consistently demonstrate poor identification of rip currents even when hazards are present (Brander et al., 2014; Castelle et al., 2016). The typical rip current structure is comprised of feeder currents, a concentrated neck, and an expanding offshore head, with maximum velocities at the neck where flow is narrowest (MacMahan et al., 2006). Rip currents are strongly influenced by coastal morphology, particularly headlands, rocky landforms that protrude into the ocean. Headlands constrain circulation currents, creating offshore flow and sea floor channels along its side, forming a persistent rip current. (Castelle et al., 2016). Rip currents are inherently complex, changing with beach characteristics and daily ocean conditions, complicating identification and hazard communication (MacMahan et al., 2006; MacMahan et al., 2010).

Dye releasing is a method to visualize rip current pathways, where researchers pour fluorescent dye into ocean feeder currents to observe its progression offshore in a rip current. Fluorescent dye is among the most effective tools for conveying rip current hazard to a public audience, because it creates a strong visual impression of the current's path (Brander et al., 2014).

Drones are being increasingly implemented for areal visualization due to their low cost and user friendly approach at data collection. Drones have been frequently used to capture and model rip current dynamics (Brander et al., 2014; Derian and Almar, 2017; Fatchurohman et al., 2021; Johansen et al., 2022). A drone dye tracing approach was applied to a known rip current system with three main objectives: 1) to assess the geospatial pattern of the rip current through a time series polygon dataset, 2) communicate these dynamics through an interactive web map, and 3) identify areas for improvement to create a set of recommendations for replicating a drone dye tracing experiment at Lovekin Rock.

2. Methods

2.1. Study Area

The study site is located at Long Beach within Pacific Rim National Park Reserve on the west coast of Vancouver Island, British Columbia, Canada. Long Beach is one of the most energetically active surf beaches on the Canadian Pacific coast. The observation area is situated at Lovekin Rock (Figure 1), a rocky outcrop along a sandy shoreline that acts as a headland rip current system. The dye release experiment was conducted under low tide conditions with moderate swell estimated at 1.5 meter wave height, sufficient for rip current circulation and aerial observation.

Data collection was designed and implemented by Mike Collyer of Parks Canada, who provided all drone footage used in this analysis. Drone operations within Pacific Rim National Park Reserve were conducted under Parks Canada Restricted Activity and Research Permits. This work was conducted alongside an ongoing rip current research partnership between the University of Victoria and Parks Canada led by Dr. Johannes Gemmrich and his graduate students Jamie Daniel and Becky Brooks.

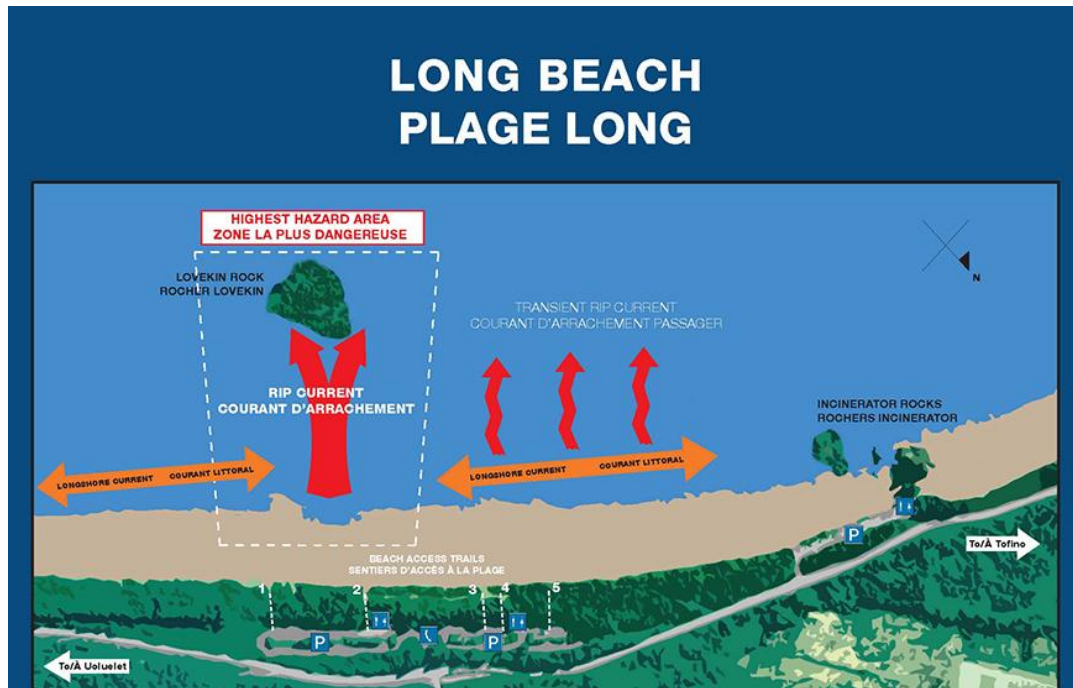


Figure 1 – Graphic from Parks Canada meant to demonstrate to visitors the longshore, and rip currents at Long Beach (Parks Canada, 2024).

2.2. Drone Data Collection

Aerial imagery was collected using a DJI Mavic 3 Enterprise multispectral drone operating at an approximate altitude of 75 m above the shoreline. The drone recorded a continuous 7 minute video capturing fluorescent dye released within the surf zone near Lovekin Rock. The camera was highly oblique, oriented at a gimbal pitch of -24.3 degrees below horizontal, approximately 66 degrees from nadir. This angle provided a wide field of view that encompassed the shoreline, the dye plume, the surf zone, and offshore waters simultaneously.

2.3. Image Extraction

14 individual images were extracted from the drone video using VLC media player (VideoLAN, 2024). Images were selected manually based on visible changes in dye plume extent, with temporal spacing ranging from 14 to 33 seconds between each frame. Images were selected where plume movement or expansion was most clearly visible and least obscured by whitewater. Dye was released 1:28 into the video, and the resulting image set spans the rest of the video.



Figure 2 – Cropped camera view ready for geoprocessing from the DJI Mavic 3 enterprise multispectral showing dye at an oblique angle. Red rectangles indicate ground control points etched into the sand and surveyed with RTK GPS.

2.4. Georeferencing

A minimum of six ground control points (GCPs) were used per image for georeferencing. Two of these were X shaped markers etched on the beach and surveyed using an RTK GPS (Figure 2, Figure 3). Four of these were derived from identifiable features where Lovekin Rock met the ocean. These were matched between the drone imagery and the Esri World Imagery basemap (Figure 3; Esri, 2022). A seventh GCP was sometimes used, based on a surfer in the ocean near Lovekin Rock equipped with an impact resistant tracking device (GPS drifter). However, this was rarely available due to limited visibility. Images were georeferenced in ArcGIS Pro using a projective transformation appropriate for oblique imagery (Esri, 2025). All imagery was georeferenced to NAD83 UTM Zone 10N. Root Mean Square Error (RMSE) forward, a measure of the average spatial discrepancy between georeferenced and actual GCP locations, was used to assess accuracy (Esri, 2024; Johansen et al., 2022).

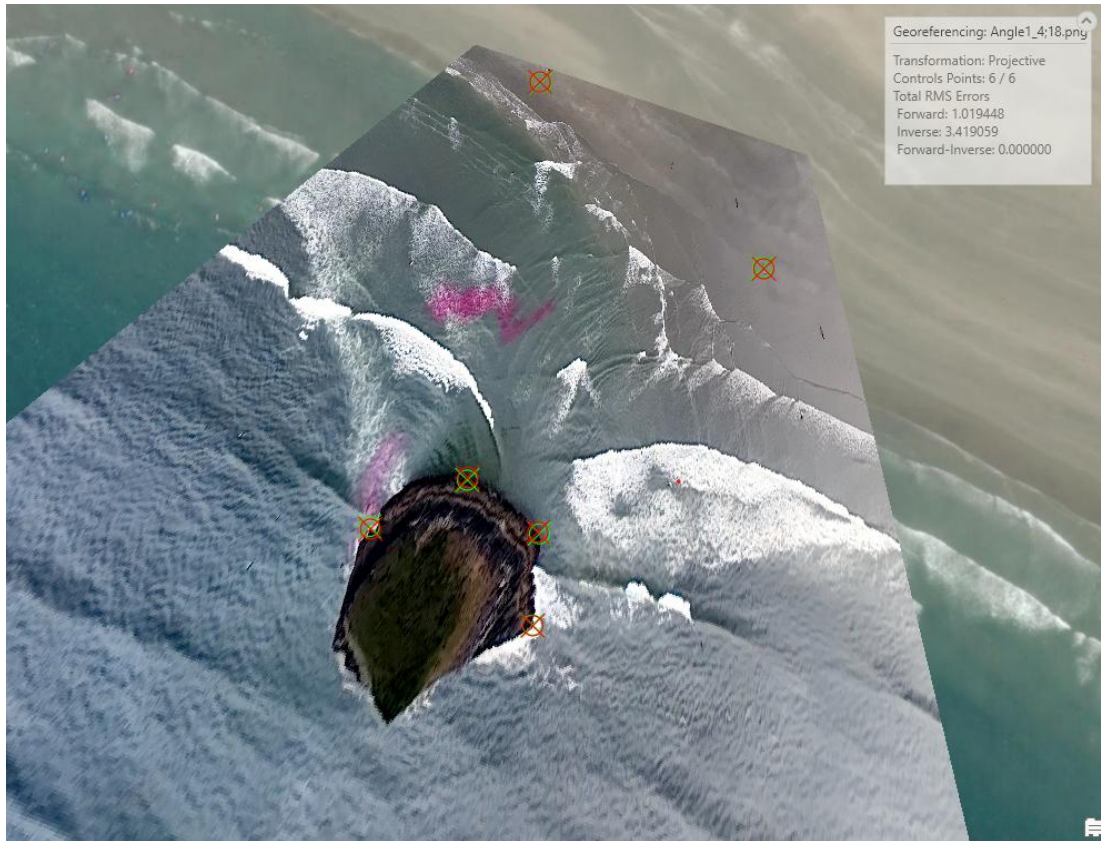


Figure 3 –ArcGIS Pro screenshot showing GCPs used for georeferencing a single image, including beach markers and the identifiable features on the rock.

2.5. Dye Plume Digitization

Following georeferencing, the visible extent of the dye plume was manually digitized in ArcGIS Pro for each image. Polygon boundaries were drawn along the outermost visible limit of the dye, where the fluorescent pink colouration transitioned to ambient water (Figure 4). Where whitewater interrupted plume connectivity, boundaries were interpolated across gaps. This produced a vector dataset of 14 polygons representing dye geometries through time and was exported as shapefiles. Attribute tables for polygons captured polygon area in m^2 , and the instantaneous rate of area change (m^2/s) was derived from the difference in successional polygons.



Figure 4 – ArcGIS screenshot of a selected polygon with 100% transparency to show the boundary outline in relation to the dye extent.

2.6. Data Processing in R

Shapefile data was imported and processed in R Studio using the `sf`, `geojsonsf`, and `jsonlite` packages (Posit Team, 2024; Pebesma, 2018; Cooley, 2020; Ooms, 2014). Timestamps from dye release was taken directly from shapefile filenames. Polygons were reprojected to WGS84 for web mapping. All plume geometries were converted from R Studio's format into GeoJSON format so they could be displayed online for web users.

2.7. Interactive Web Map

An interactive web map was developed using the `leaflet` package and rendered within a Quarto Hypertext Markup Language (HTML) document styled with custom Cascading Style Sheets (CSS) (Cheng et al., 2023; Allaire et al., 2022). The map is centred on the spatial extent of the plume data and uses an Esri World Imagery basemap (Esri, 2022). Rather than using the leaflet software's built in layer switching tools, a custom interactive control panel was developed and overlaid directly on the map with JavaScript and html widgets (Vaidyanathan et al., 2023). The panel contains a manual timeline slider and a play/pause button that animates the polygon sequence at 0.7 second intervals. At each timestep, the corresponding dye polygon is displayed on the map and the previous removed. A statistics panel below the map updates alongside the slider, displaying

plume area, rate of change, colour coded green for expansion and red for contraction, and the current timestep out of the total sequence. All data, including spatial geometries, time labels, area values, and rates of change, were bundled into a single dataset in R Studio and passed to the map interface at the time of rendering. This avoids the need for any external data requests during usage. Attached is the website where dye progression can be viewed, <https://emmettvannier.github.io/LongBeachDyeRelease/Emmett.html>.

3. Results

Georeferenced images had RMSE forward values ranging from approximately 1.02 to 5.39 metres with an average of 3.22 across the image set. Polygon area ranged from 142.5 m² to 2295.4 m². Polygon rate of change ranged from -14.21 m/s to +29.19 m/s. The dye plume exhibited multidirectional transport in the early stages of the release as initial dispersion was diffuse, with dye moving both alongshore under the influence of a longshore current, onshore via wave energy, and offshore under the influence of the rip (Figure 5). As the dye progressed, clear offshore transport became evident, as the dye progressively concentrated toward Lovekin Rock. Dye progression is available for observation on the webpage.



Figure 5 – Three progressive snapshots of dye progression from the webpage. From left to right are the dyes at 00:36 after dye release, 3:42 after dye release, and 5:25 after dye release.

4. Discussion

Assessing the dye's progression provides geospatial evidence of the headland rip current at Lovekin Rock. The longshore component of transport in early time steps indicates influence from the longshore current at Long Beach (Figure 1). Then, the transition from diffuse spread to channelization towards Lovekin Rock reflects the feeder to neck progression of a headland rip current (MacMahan et al., 2006). Fluorescent dye as a rip current visualization tool was effective to observe this process. Headland rip currents are anchored by fixed morphology rather than migrating sandbars that control transient currents (Castelle et al., 2016). This persistent, predictable character distinguishes the Lovekin Rock rip current from more variable rip currents and has direct implications for hazard management within the Pacific Rim National Park Reserve. The current enables targeted interventions such as the seasonal deployment of the Parks

Canada Coastal Steward program to this location. Stewards educate beachgoers about the risks of entering the area without experience and intervene when inexperienced beach users enter or are caught in the rip current. Long Beach is nationally significant, and the web tool developed may provide enhanced rip current education for visitor safety.

4.1. Limitations

Georeferencing oblique camera geometry introduces distortion as it stretches a tilted image to fit a flat map. A mathematical correction is applied to account for the distortion but weakens in areas further from GCPs (Verykokou & Ioannidis, 2018). GCPs in this experiment were limited and unevenly distributed across the drone's field of view. GCPs in the ocean are difficult to collect, making distributing GCPs evenly across the frame a challenge.

The quality of RTK GCPs points was high. The four GCPs taken on Lovekin Rock features have human error inherent with manual identification that is hard to quantify as it relies on alignment of features between the drone and satellite imagery. Rock features can be inconsistent as tides and waves fluctuate between images, shifting where the rock meets the ocean. Additionally, these GCPs were identified with the ESRI World Imagery basemap which has lower accuracy in remote areas (ESRI, 2022). The strongest GCPs were on the beach while the four weakest ones were on the rock where georeferencing accuracy is most important for measuring the rip current.

Fluorescent dye is a passive tracer, and manual digitizing introduces a source of uncertainty that is difficult to quantify. The plume boundary was defined by a visual threshold, and breaking waves obscured portions of the dye, demanding interpolation across breaks in visibility. Progressive dilution also reduced plume visibility over time, introducing a temporal accuracy bias toward earlier, higher concentration images (Johansen et al., 2022; Fatchurohman et al., 2021). Buoyancy effects may cause the dye to decouple from the bulk water column and better represent surface flows. This means the dye may not perfectly represent the bulk water movement beneath it, which is integral to the current's circulation (Dalrymple et al., 2011).

4.2. Recommendations

The following steps are recommended for replicating a drone dye tracing experiment at Lovekin Rock:

Drone altitude and camera angle. Fly at the maximum permitted altitude with the camera pointed straight down. Important to note Long Beach's proximity to Tofino Airport requires advance coordination with Transport Canada for airspace authorization and flight altitude will be subject to permission. If the permitted flight altitude is insufficient for capturing the experiment at a nadir angle, multiple drones could be implemented to cover the experiment.

Ground control points. Place a minimum of ten clearly visible GCPs per image and distribute them across as much of the image as possible. GCPs etched in the sand should be uniquely identifiable from the drone imagery, such as adding a number in the sand next to each marker. GCPs should be placed on Lovekin Rock to provide accurate control points in this portion of the

frame. Additionally, more surfers with GPS drifter equipment should be implemented, and surfers should wear high visibility clothing distinguishable from each other, i.e. different colours. GCPs should be taken with precision equipment i.e. GPS drifters and RTK GPS when possible.

Image capture. Capture still images with RGB sensor at consistent 5–15 second intervals rather than extracting frames from video. Still images will produce higher resolution output and remove the need for manual frame selection.

Timing and conditions. Conduct the experiment during a low tide with wave height exceeding 1m, in low winds, and within two hours of solar noon to minimize sun glint on the water surface.

Dye release. Release large volumes of dye to maximize plume visibility throughout the full observation window.

5. Conclusion

Geospatial analysis assessed the spatial evolution of a fluorescent dye plume released into the Lovekin Rock headland rip current system at Long Beach, British Columbia. The observed plume trajectory was consistent with headland rip current structure and confirms the persistent nature of the Lovekin Rock hazard. An interactive web map communicated this flow pathway, offering a platform for scientific reporting and public hazard education within Pacific Rim National Park Reserve.

The workflow demonstrated that time series dye plume digitization from drone video provided a practically informative approach to rip current visualization, even under the constraints of oblique imagery and limited GCPs. Future work should prioritize nadir drone imagery and increase high quality GCP coverage. As drone technology and coastal monitoring practice continue to develop, drone dye tracing prevails as a cost effective and accessible tool to assess rip current dynamics and communicate coastal hazards.

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